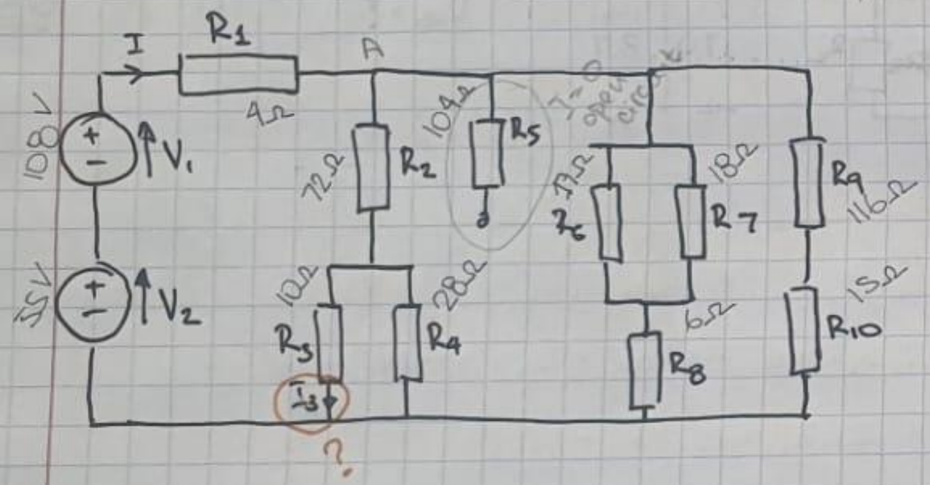
② Practice

$$R_G = \frac{1}{1/96595 + 1/1.18\times10^6} = 89.286\ k\Omega$$

$$V_{@R6} \Rightarrow \frac{89.286\times10^3}{89286+90933+23+1.6}\times238 = 117.896\ V$$

$$\text{ERROR} = \frac{117.896}{-122.577} / 122.577 = -3.819\%$$

Q5



KCL @ A

$$\frac{V_a - V_1 - V_2}{R_1} + \frac{V_a}{R_2 + 1/(1/R_3 + 1/R_4)} + \frac{V_a}{R_8 + 1/(1/R_6 + 1/R_7)} + \frac{V_a}{R_9 + R_{10}} = 0$$

$$\frac{V_a - 108 - 55}{4} + \frac{V_a}{72 + 1/(1/10 + 1/28)} + \frac{V_a}{6 + 1/(1/7 + 1/8)} + \frac{V_a}{116 + 15} = 0$$

$$\frac{V_a - 163}{4} + \frac{19\,V_a}{1508} + \frac{35\,V_a}{516} + \frac{V_a}{131} = 0$$

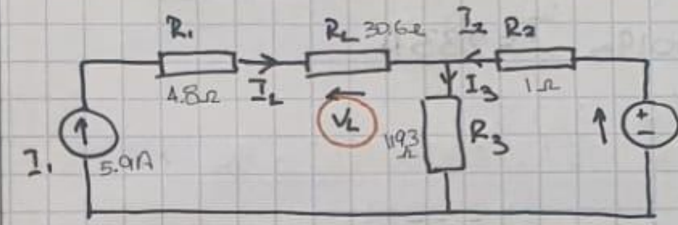
$$-163/4 + 0.338\,V_a = 0 \Rightarrow V_a = 120.56\ V$$

apply current division Rule

$$I_2 = \frac{V_a}{R_2 + 1/(1/R_3 + 1/R_4)} \qquad I_3 = \frac{I_2 R_4}{R_3 + R_4}$$

$$I_2 = \frac{120.56\times19}{1508} = 1.519 \Rightarrow I_3 = \frac{1.519\times28}{28+10} = 1.119\ A$$

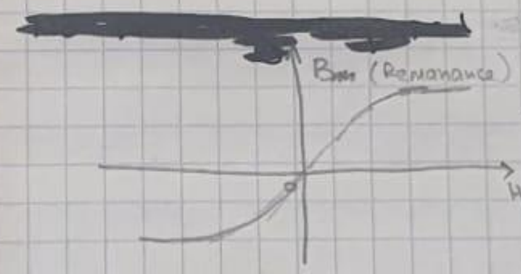
Q6



V = IR

$I_L = I_1 = 5.9\ A \Rightarrow V_L = 30.6\times5.9 = -180.54\ V$

Q7



as Remanance is approached the flux becomes stationary as gradient no longer varies.

therefore I would remain relatively constant & unchanged

Q8

L = 293mm V = 13.7 m/s V_{EMF} = 0.5V F_m = 98.2N

G = ?

$V_{emf} = uBL \Rightarrow 13.7\times0.293\times B = 0.5 \Rightarrow B = 0.125\ Wb/m^2$

$F_m = ILB \Rightarrow 98.2 = I\times0.293\times0.125 \Rightarrow I = 2681.229\ A$

$R = \frac{V_{emf}}{I} = \frac{0.5}{2681.229} = 0.000186\ \Omega \Rightarrow G = 1/R = 5362.457\ S$